

Laplace method for computing det

$A = \begin{bmatrix} 0 & 1 & 3 \\ 4 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$

1) Pick any column or row
 2) $0 \begin{vmatrix} 0 & 2 \\ 1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 4 & 2 \\ 0 & 1 \end{vmatrix} + (-3) \begin{vmatrix} 4 & 0 \\ 0 & 1 \end{vmatrix} = -16$

matrix from removing top row left column
 remove top row middle column
 remove top row right column

coefficient alternate signage $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$

$$\begin{vmatrix} 0 & 1 & -3 \\ 4 & 0 & 2 \\ 0 & 1 & 1 \end{vmatrix} = -1 \begin{vmatrix} 4 & 2 \\ 0 & 1 \end{vmatrix} + 0 \begin{vmatrix} 0 & 2 \\ 0 & 1 \end{vmatrix} - 1 \begin{vmatrix} 0 & -3 \\ 4 & 2 \end{vmatrix} = -4 - 12 = -16$$

EX / Laplace for 2×2 $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = a|d| - b|c|$

best choice is c, our 3 minors

$$\begin{vmatrix} 0 & 1 & -3 \\ 4 & 0 & 2 \\ 0 & 1 & 1 \end{vmatrix} = 0 \begin{vmatrix} 4 & 2 \\ 0 & 1 \end{vmatrix} - 4 \begin{vmatrix} 1 & -3 \\ 1 & 1 \end{vmatrix} + 0 \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = -16$$

EX: 4×4

$$\begin{vmatrix} 1 & 0 & 6 & -1 \\ 2 & -1 & 3 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{vmatrix} = 0 \begin{vmatrix} 2 & 3 & 0 \\ 0 & 1 & 2 \end{vmatrix} + 1 \begin{vmatrix} 1 & 6 & -1 \\ 2 & 3 & 0 \\ 0 & 1 & 2 \end{vmatrix} - 0 \begin{vmatrix} 1 & 0 & 6 \\ 2 & -1 & 3 \\ 0 & 1 & 1 \end{vmatrix} + 1 \begin{vmatrix} 1 & 0 & 6 \\ 2 & -1 & 3 \\ 0 & 1 & 1 \end{vmatrix}$$

$$= 0 - 1 \begin{vmatrix} 1 & -1 \\ 2 & 0 \end{vmatrix} + 2 \begin{vmatrix} 1 & 6 \\ 2 & 3 \end{vmatrix} + 0 - 1 \begin{vmatrix} 1 & 6 \\ 2 & 3 \end{vmatrix} + 1 \begin{vmatrix} 1 & 6 \\ 2 & -1 \end{vmatrix}$$

$$= -2 + 2(-9) - (-9) + (-1) = -12$$